

Revision Note

Paper: Receiver-Domain Behavioral Probing for Backdoor-Resilient Federated GPS Spoofing Detection

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This was a rebuild around one story, not a compression. Nothing was shrunk to fit; sections were removed, merged or rewritten based on whether they advance the argument a reviewer has to be convinced by.

Result: 15 pages, 12 tables/figures, 25 subsections becomes ~6 pages, 1 table and 4 figures, 12 references. The title changed to lead with the mechanism rather than the property, since the mechanism is the contribution.

The story the paper now tells

Self-reported performance is an exploitable attack lever in federated UAV spoofing detection. We propose receiver-domain behavioral probing, which lets the coordinator judge client models on evidence it generates itself. Four result elements carry the whole argument, in order:

attack validity -> competitive comparison -> key differentiator -> limitation and generalization

Everything that did not serve that chain was cut.

Retained, and why

Element	Why it stays
Table I (8-row comparison)	Proves the attack works, inflation makes it worse, median and FLTrust leave residual lift, Multi-Krum is competitive, and ours attributes per client
Fig. 2 (unknown attacker count)	The strongest differentiator: existing rules need f, we do not
Fig. 3 (heterogeneity)	The honest boundary, and the empirical justification for keeping the median backstop
Fig. 4 (trigger generalization)	Supports the scoped claim that the trigger feature need not be known
Fig. 1 (system/threat model)	The authors' own diagram, showing the three attack steps and the inflated weight path
Algorithm 1	Shortened from 20 lines to 11; the method is the technical center

Removed from the main paper

All of this remains in the repository artifact and is summarized in one or two sentences where it matters.

- **Positioning table against related work** -- replaced by two compact paragraphs, per instruction
- **Layer-ablation table** -- subsumed by Table I, which contains the same conditions
- **Hyperparameter sweep** table and figure -- one sentence in §V-E
- **Trust-degradation table** -- one clause; its decisive case is now Fig. 3
- **Centralized detector-ceiling table** -- two clauses in the Discussion limitations
- **Cost scaling plots** and the two-denominator table -- two sentences in §V-E
- **Adaptive-attacker table and figure** -- three sentences in §V-E
- **Feature-separability table** (10 rows) -- replaced by the phrase "eight satisfy $d \geq 0.05$ "
- **False-positive breakdown table** -- folded into the 0.3% figure in Table I and §V-A
- **Failed Dirichlet experiment** -- cut entirely; the working partition is described in one sentence
- **Preprocessing table** and per-step row counts -- reduced to two sentences
- **Round-progression, sensitivity and FLTrust-only figures** -- the tables carry the same numbers

Rewritten rather than trimmed

- **Abstract** : 340 -> 192 words. Problem, attack, idea, one attack number, one defense number, one differentiator, one scope sentence. Multi-Krum stays because it frames the differentiator; the adaptive attacker, cost, trigger sweep and false-positive rate are all out.

- **Introduction** : ~1,900 -> 617 words. Related work compressed from four paragraphs of per-paper summary into one gap paragraph and one FLTrust-positioning paragraph. Contributions cut from five bullets to three.

- **System and threat model** : the three attack steps are now labeled A1-A3 and expressed with symbols, with one equation each where it earns one.

- **Results** : reorganised so each subsection maps to exactly one of the four elements.

- **Discussion and Conclusion** : merged into one section.

References

12, down from 12 -- the count was already in range, but two were re-verified against source during the previous revision and the dataset citation was corrected to the Mendeley *Unmanned Aerial System* release (`10.17632/z7dj3yyzt8.3`), which matches the data actually used. The bibliography is now inline (`thebibliography`) rather than BibTeX, so the paper compiles in one pass with no `.bbl` step.

Figures

Figures 2 to 4 rebuilt at publication quality and exported as **600 dpi PNG** , twice the 300 dpi IEEE asks for line art:

- single IEEE column width (3.4 in) so none needs a full-width float
- 6.5 to 8.5 pt type, legible at print size
- consistent colour role across figures (red = attack/undefended, green = proposed, purple = Multi-Krum, blue = FLTrust, grey = median)

Figure 1 is the authors' own diagram rather than a generated one. It shows the three attack steps and, importantly, the path by which the inflated self-reported accuracy drives the aggregation weight toward its maximum, which is the specific vulnerability the paper targets.

One thing we did not do

The paper does **not** claim the proposed method dominates every baseline, because our own Table I would contradict it. Multi-Krum is statistically indistinguishable from behavioral trust alone under the nominal IID setting and costs far less server time. That result is stated plainly in the abstract, in §V-A and in the Conclusion, and is then used to motivate the two properties that do distinguish the method: no attacker-count parameter, and per-client attribution.

Deliverables in this folder

File	What it is
<code>`main.tex`</code>	The paper, IEEEtran `conference` class, inline bibliography
<code>`figures/fig2..4_*.png`</code>	Figs. 2-4 at 600 dpi; Fig. 1 is supplied by the authors
<code>`build_figures.py`</code>	Regenerates Figs. 2-4 from the CSVs
<code>`14_conference_results.ipynb`</code>	Executed notebook reproducing Table I and Figs. 2-4
<code>`fl_common.py`, `fl_runner.py`</code>	The shared harness (split, model, attack, 9 aggregation rules, metrics)
<code>`exp_*.py`, `run_all.py`</code>	The experiments behind the reported results
<code>`results/*.csv`</code>	Every reported value, exported

Reproduce with `python run_all.py baselines attackers noniid` then `python build_figures.py` , or set `RUN_EXPERIMENTS = True` in the notebook.